

AMANDEEP SINGH

AI Software Developer | LangGraph • Node.js • FastAPI • Full Stack

Ferozpur, Punjab | +91 94648 20510 | amandeep2975@gmail.com
GitHub: <https://github.com/Aman2975>
Portfolio: <https://iaman.space/>
LinkedIn: <http://www.linkedin.com/in/amandeep2975>

SUMMARY

Software Developer specializing in backend development with hands-on experience building production-grade systems. Built ShodhKartaChecker (AI-powered plagiarism detection engine) using Node.js, FastAPI, Redis task queues, and Groq AI, achieving 40% faster API response times. Skilled in microservices architecture, JWT authentication, RBAC, and REST API design. Looking to contribute to a software engineering team and solve real-world engineering problems.

SKILLS

- Languages: JavaScript (Node.js), Python, TypeScript
- Backend: REST APIs ,Microservices ,JWT Authentication ,RBAC ,Swagger ,Async Processing ,Caching Redis Queues , AI API Integration
- Frameworks: Express.js, React.js, NestJS, FastAPI, LangChain, LangGraph
- Databases: MySQL, MongoDB, Redis, PostgreSQL
- AI & Agents: LangChain, LangGraph, AI Agents, Tool Calling, NLP, Groq AI, LLM Integration, Agentic Workflows, Natural Language Processing
- ORMs: Prisma , Mongoose , TypeORM
- DevOps: Docker , Nginx , PM2 , Linux , Git , GitHub, CI/CD
- Testing: Jest, Supertest , Postman
- Cloud & Storage: Cloudinary , MongoDB Atlas
- Deployment: VPS, Vercel, Render
- Concepts: System Design , Data Structures, DB Design, Error Handling, MVC Architecture, Backend Optimization, Asynchronous Processing, Concurrency

CERTIFICATION AND TRAINING

Software Development Training – Department of Computer Science Punjabi University, Patiala

(MCA Industrial Training Program)

Duration: 1 Jan 2026 – 5 May 2026

- Built an AI-Powered Plagiarism Detection System with Redis-based task processing and Groq AI API integration

PROJECTS

AI-First CRM for Healthcare Professionals | [GitHub](#)

FastAPI • Python • LangGraph • LangChain • Groq AI • PostgreSQL • React

- Built an AI-first CRM where pharmaceutical reps describe doctor meetings in natural language — AI extracts structured data and auto-fills CRM forms eliminating manual entry
- Designed LangGraph workflow orchestrating 5 business tools — interaction logging, editing, search, follow-up scheduling, and AI recommendation generation

- Integrated LangChain + Groq Gemma2-9B for intent detection, structured output extraction, and conversational AI responses
- Built React + Redux Toolkit frontend with real-time chat interface synced to structured CRM form via shared application state
- Applied modular monolithic architecture with clean separation across API, workflow, AI, business, and persistence layers

ShodhKartaChecker (AI-Powered Plagiarism Detection System) | [GitHub](#)

Node.js . Python . Express.js . FastAPI . MongoDB . Redis . Groq AI . Docker

- Built a distributed plagiarism detection platform integrating **2 backend services** (Node.js and FastAPI).
- Implemented **Redis-based asynchronous task queues**, reducing API response time by **40%**.
- Architected a **microservices-based** workflow for **document upload, preprocessing, AI analysis, and report generation**.
- Integrated **Groq AI APIs** for plagiarism explanations.
- Deployed the application using **Docker, Nginx, and PM2** on a VPS.

PostGrid (Blog Platform) | [GitHub](#)

Node.js • Express.js + Typescript • PostgreSQL • Multer • PrismaORM

- Developed **15+ REST API endpoints** supporting CRUD operations, authentication, and profile management.
- Integrated Cloudinary for cloud media storage and Multer for multipart file uploads.
- Implemented a PostgreSQL database with **3+ relational models** using Prisma ORM.
- Containerized the application using Docker for consistent deployment.
- Implemented JWT authentication for secure user access.

EDUCATION

Master of Computer Applications (MCA)

Punjabi University – Patiala, India

Completed: May 2026

CGPA:8.1